

1.1.2 Introduction to Digital Systems

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بٹ (Bit) کہتے ہیں۔

CS XI

Chapter 1: Computer Systems

Lecture 02 of 20

1.1.2 Introduction to Digital Systems

LEARN IT

A **digital system** is an electronic system that uses only two values: **0** and **1**. These two values are called **binary digits**, or **bits** for short.

- **0** represents OFF, LOW, or FALSE.
- **1** represents ON, HIGH, or TRUE.

Why only two values? Because it is easy for an electronic circuit to tell "voltage present" from "no voltage". Every number, letter, photo and song in a computer is finally stored as a long pattern of these bits.

Advantages of digital systems

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Examples of digital systems

- **Computers:** process data as 0s and 1s.
- **Smartphones:** store photos and music as binary data.
- **Digital cameras:** capture images as binary code.

WORKED EXAMPLE

Why is copying digital data "accurate"?

Step 1: A song is stored as bits, for example 10110010...

Step 2: When you copy it to a USB, the computer copies each bit exactly: 1 stays 1, 0 stays 0.

Step 3: Even after 100 copies the bits are identical, so the song sounds the same. An old cassette tape (analog) loses a little quality every time it is copied.

REMEMBER IT

Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

اردو میں سمجھیں

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بٹ (Bit) کہتے ہیں۔ 0 کا مطلب بند، کم یا غلط ہے اور 1 کا مطلب چالو، زیادہ یا درست ہے۔ ڈیجیٹل نظام قابل بھروسا، درست، بنانے میں آسان اور سافٹ ویئر سے قابو میں رہنے والے ہوتے ہیں۔

CHECK YOURSELF

1. What does a 1 represent in a digital system?
2. Give two advantages of digital systems.

Answers: 1. ON, HIGH or TRUE. 2. Any two of: reliable (less affected by noise), accurate (no loss when copying), easy to design (two states), programmable (controlled by software).

Next lecture: 1.1.3 Analog and Digital Signals